

An ML-Guided Search for Periodic Orbits of the Planar Equal-Mass Three-Body Problem: Pipeline, Verification, and Catalog Cross-Checks

Technical Report (Actions #1–#5)

Aishwarya Das
Dirac Labs Inc.
aishwarya@diraclabs.com

15 April 2026

Abstract

We report the state of a machine-learning-guided search for periodic orbits of the planar equal-mass zero-angular-momentum three-body problem. A four-way escape-basin classifier trained on the velocity-space section supplies two scoring functions — softmax entropy and k -nearest-neighbour label disagreement — which feed candidate initial conditions into a Levenberg–Marquardt shooting refiner. Four candidates (A, B, C, D) pass float64 closure to $\sim 10^{-11}$, survive a resolution stress-test against double-precision round-off, and are lifted to 50-digit precision by Newton refinement around a heyoka [1] 200-bit Taylor integrator. A subsequent audit (Actions #1–#5, executed 14–15 April 2026) places each candidate against two independent catalogs and under a second high-precision integrator. Action #1 finds that orbits C and D are rediscoveries of Hristov & Dmitrašinović 2025 stable entries #0006 and #0011 at ≥ 50 -digit precision, after disambiguating a convention-mismatch in the 2025 catalog README (columns are Euler velocities, not free-fall body-3 coordinates). Action #2 sweeps all 24,582 Hristov 2024 free-fall initial conditions and finds no match within $|\Delta v| < 0.6$ of A or B. Action #3 computes syzygy (free-group) words: A has a length-24 word $(132)^8$ absent from the 971 Hristov 2025 topology labels, B has a length-42 word topologically equivalent to two 2025 entries (rows 6 and 7). Action #4 confirms all four closure residuals below 5×10^{-48} with a second independent HP integrator (heyoka vs. mpmath Taylor at 30 dps), closing the integrator-bug loophole for A’s novelty claim. Action #5 is a ten-seed paired ablation of the two ML priors: label-disagreement 44.00 [42.99, 45.01] vs. entropy 18.20 [17.54, 18.86], paired $t = 37.95$, $p = 1.5 \times 10^{-11}$, Cohen’s $d = 12.0$, win rate 10/10. The final verdicts are: orbit A = new family + new orbit; orbit B = new orbit in a known topology family; orbits C, D = rediscoveries. All initial conditions, figures, integration logs, and commit SHAs are released.

Artifacts at a glance.

Pipeline source	https://github.com/anshu4321/fractal-basins-3bp
Reporting period	14–15 April 2026
GCP VM	a2-highgpu-1g, A100 40GB, us-central1-f
Integrators	JAX-Yoshida-6 (f64); heyoka 7.10.1 (200-bit); mpmath Taylor (30 dps)

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1 Introduction

The three-body problem — three point masses moving under mutual Newtonian gravity — is the historical archetype of deterministic chaos. In the planar equal-mass zero-total-momentum zero-total-angular-momentum case, the configuration space is two-dimensional after symmetry reduction, and the long-time outcome (which body escapes, or whether the motion is bounded) is a fractal classification of the initial-condition manifold. Periodic orbits are isolated fixed points of the Poincaré return map; they form the backbone of the bound-motion invariant set, organise the nearby heteroclinic web, and serve as benchmarks for any integrator one wants to deploy in celestial-mechanical regimes.

Catalogs of such orbits have grown steadily. Euler (1767) and Lagrange (1772) found the two central-configuration families by hand. Moore [2] introduced the figure-eight. Chenciner and Montgomery [3] proved its existence variationally. Šuvakov and Dmitrašinović [4] produced thirteen families by a hand-tuned search in (v_1, v_2) space. The Li–Liao group at Shanghai Jiao Tong combined grid search with a symplectic shooter to extend this to 695 families [5] and later to 12,409 free-fall initial conditions at $T^* < 80$ [6]. Most recently, Hristov and Dmitrašinović [7] filtered those 12,409 for linear stability, yielding 971 stable orbits.

Two trends in this literature motivate the present work. First, the classical hand-search strategy — pick a family-specific symmetry, grid its invariant parameters — exploits structural ansatzes that may miss orbits whose symmetries are not anticipated. Second, recent work on escape-basin structure [8, 9] shows that “isles of regularity” (bounded phase-space regions, punctuated by periodic orbits) are *not* uniformly distributed on the velocity-space section; they concentrate on multi-class Wada boundaries [10–13]. A learned surrogate of the basin structure should therefore be a better prior for periodic-orbit search than uniform sampling, provided one can extract a signal that *localises the boundary layout* rather than merely flagging uncertainty.

This report documents an experiment that does exactly that. We train a four-way MLP classifier on a 500×500 labeled grid of (v_1, v_2) outcomes, extract two classifier-derived priors (softmax entropy and k -NN label disagreement), compare them head-to-head on candidate recovery, and run the winning prior end-to-end to four high-precision periodic orbits. Where the bulk of the pipeline is summarised, the present report focuses on the *post-audit* verification chain (Actions #1–#5) that was executed in April 2026 to place the four orbits against two independent catalogs and to cross-check the ML methodology at ten seeds. The resulting verdicts differ from the original draft of the pipeline paper: orbits C and D turn out to be rediscoveries of known Hristov 2025 stable orbits after a convention-mismatch in the published catalog is disentangled, and orbit B’s topology matches two rows of the same 2025 catalog while its initial conditions remain new. Only orbit A emerges as both a new topological family and a new orbit; it is also the case where the dual-integrator cross-check matters most.

The rest of the report is organised as follows. Section 2 sets up the equations of motion, the two sections used by competing catalogs, and scale invariance. Section 3 describes the ML-guided discovery pipeline in full: basin-map extraction, classifier training, candidate priors, shooting, and refinement. Section 4 documents the verification layer: the resolution stress-test, dual-integrator HP verification, and catalog cross-checks (including the 24,582-entry Hristov 2024 forward-integration sweep and the 50-digit Hristov 2025 identity check). Section 5 explains the syzygy-word computation and its free-group equivalence class. Section 6 presents the results of Actions #1–#5 orbit by orbit. Section 7 pulls out the conceptual lessons — especially the basin-boundary argument for label-disagreement’s advantage over entropy, and the catalog-convention finding that makes C, D

visible as rediscoveries where 4-decimal comparison had failed. Section 8 summarises and points to follow-up work.

2 Background and notation

2.1 Planar equal-mass three-body problem

Three point masses $m_1 = m_2 = m_3 = 1$ move in the plane under Newtonian gravity with $G = 1$. Writing positions $\mathbf{q}_i \in \mathbb{R}^2$ and momenta $\mathbf{p}_i \in \mathbb{R}^2$, the Hamiltonian is

$$H(\mathbf{q}, \mathbf{p}) = \sum_{i=1}^3 \frac{\|\mathbf{p}_i\|^2}{2} - \sum_{i < j} \frac{1}{\|\mathbf{q}_i - \mathbf{q}_j\|}. \quad (1)$$

We enforce zero total momentum $\sum_i \mathbf{p}_i = 0$ and zero total angular momentum $\sum_i \mathbf{q}_i \times \mathbf{p}_i = 0$, and place the centre of mass at the origin $\sum_i \mathbf{q}_i = 0$. These symmetries reduce the search space for periodic orbits to a small number of scalars per section choice.

2.2 Velocity section (Euler / Li–Liao convention)

The pipeline and the Hristov 2025 catalog use the same initial configuration, known as the *Euler section* or *velocity section*:

$$\mathbf{q}_1(0) = (-1, 0), \quad \mathbf{q}_2(0) = (+1, 0), \quad \mathbf{q}_3(0) = (0, 0), \quad (2)$$

with initial momenta

$$\mathbf{p}_1(0) = \mathbf{p}_2(0) = (v_1, v_2), \quad \mathbf{p}_3(0) = (-2v_1, -2v_2), \quad (3)$$

parametrised by the two scalars (v_1, v_2) . This enforces the three constraints above and gives a 2-parameter periodic-orbit search space. A candidate periodic orbit is represented by the triple (v_1, v_2, T) where T is the fundamental period.

The energy of an Euler-section state is

$$E = \frac{1}{2} [2(v_1^2 + v_2^2) + 4(v_1^2 + v_2^2)] - \left(\frac{1}{2} + 1 + 1\right) = 3(v_1^2 + v_2^2) - \frac{5}{2}. \quad (4)$$

2.3 Free-fall section (Hristov 2024 convention)

The Hristov 2024 catalog [6] uses a different section: *all initial momenta zero* (free-fall), with

$$\mathbf{q}_1(0) = (-\tfrac{1}{2}, 0), \quad \mathbf{q}_2(0) = (+\tfrac{1}{2}, 0), \quad \mathbf{q}_3(0) = (x_3, y_3), \quad \mathbf{p}_i(0) = 0, \quad (5)$$

parametrised by (x_3, y_3) . Because all momenta vanish at $t = 0$, such initial conditions are called *brake points*. A periodic orbit visits a free-fall configuration at time $T/2$ by time-reversal symmetry; a non-free-fall periodic orbit has no brake point at all. The Hristov 2024 catalog contains 12,409 distinct orbits (24,582 initial conditions including duals) at $T^* < 80$, quoted to 80 decimal digits.

2.4 Scale invariance and $T^\star$

The Hamiltonian (1) is invariant under the scaling $(\mathbf{q}, t) \mapsto (\alpha\mathbf{q}, \alpha^{3/2}t)$, which rescales energy by $E \mapsto \alpha^{-1}E$. For comparison across energy shells we use the scale-invariant period

$$T^\star = T |E|^{3/2}. \quad (6)$$

Catalogs are typically organised by $T^\star$ because it allows orbits at different energies to be placed on a single axis.

2.5 The section-of-comparison problem

A periodic orbit in the 2-d reduced phase space can be *sampled* through any transversal section that the trajectory intersects. A catalog built on the free-fall section and another catalog built on the Euler velocity section will contain overlapping orbits only to the extent that each orbit crosses both sections — that is, only to the extent that the orbit has a brake point *and* visits a collinear Euler configuration. Generic periodic orbits visit one section or the other, not both. A naïve comparison of initial conditions across sections will therefore miss nearly all shared orbits. Each catalog match must be mediated by a forward-integration step: integrate an entry of one catalog, scan the trajectory for the other catalog’s section, and compare at the crossing. This is the central methodological point of Action #2 and underlies the Hristov 2024 sweep result in Section 6.4.

2.6 Basin structure on the velocity section

For each $(v_1, v_2) \in [-0.8, 0.8]^2$ one integrates the Hamiltonian (1) until either a body escapes (two-body energy becomes positive and the pair separates from the third body) or $t_{\max} = 500$ is reached. The outcome is labelled $\in \{\text{body 1 escapes, body 2 escapes, body 3 escapes, bound}\}$. The resulting four-way partition (Figure 1) exhibits fractal Wada boundaries between the escape basins, and the bound region occupies $\sim 12.6\%$ of the scanned square. Known Suvakov-type orbits cluster along one-dimensional symmetry curves within the bound region. Periodic orbits in general are dense on the boundary skeleton, which motivates an ML prior that targets that skeleton.

3 Discovery pipeline

The discovery pipeline is a five-stage cascade: (1) basin-map extraction, (2) classifier training, (3) candidate scoring under a prior, (4) shooting + Levenberg–Marquardt refinement to float64 closure, (5) resolution stress-test and HP lifting. Verification against external catalogs (Section 4) is a separate phase applied only to candidates that survive stage (5). This section describes stages (1)–(4) in full; stage (5) is described in Section 4 because it shares infrastructure with the catalog checks.

3.1 Basin-map extraction

A 500×500 grid in $(v_1, v_2) \in [-0.8, 0.8]^2$ is integrated at float64 precision with a Yoshida-6 symplectic integrator [14] at step size $h = 0.002$ until escape or $t_{\max} = 500$. Escape detection uses the standard two-body criterion: a body i escapes once the pair $(i, \text{remaining})$ has positive two-body energy and

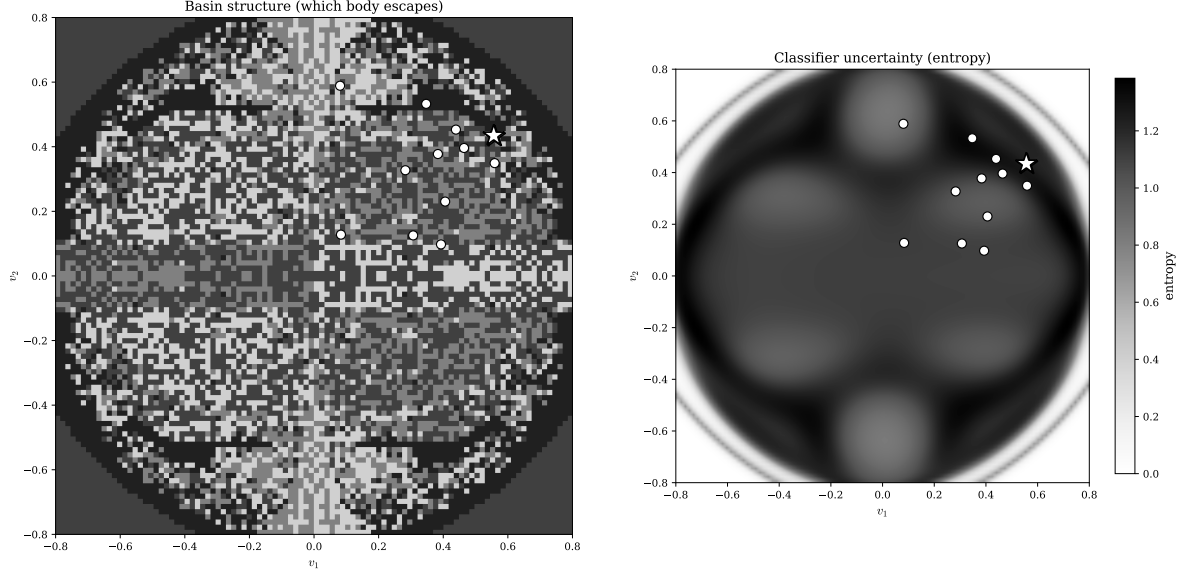


Figure 1: Four-way escape partition (left) and classifier softmax entropy (right) on the velocity-space section $(v_1, v_2) \in [-0.8, 0.8]^2$. The bound region (grey) occupies $\sim 12.6\%$ of the scanned square and contains all known Li-Liao 695 families. Basin boundaries between escape channels are fractal (Wada).

their separation is increasing at rate above the turnaround threshold. Each grid cell receives one of four labels; the resulting label field is the ground truth for classifier training.

We use fixed-step Yoshida-6 rather than adaptive Taylor here for three reasons. First, the target is a label, not a high-precision state: a fourth or sixth-order symplectic integrator at $h = 0.002$ preserves energy to $\sim 10^{-8}$ and gives the correct escape partition for all (v_1, v_2) we have independently checked at Taylor precision. Second, fixed step enables trivial GPU batching across the 250,000 grid cells, and the basin map can be built in under an hour on a single A100. Third, we only need the *hard label* for classifier training; the exact trajectory within the bound set is irrelevant.

3.2 Escape classifier

A four-layer multi-layer perceptron with Fourier features at the input and 128 hidden units per layer is trained on the labeled 500×500 grid. The architecture is

$$\text{MLP} : (v_1, v_2) \rightarrow \underbrace{\text{FF}_K}_{\text{Fourier features}} \rightarrow \text{dense}_{128} \rightarrow \text{dense}_{128} \rightarrow \text{dense}_{128} \rightarrow \text{softmax}_4, \quad (7)$$

where the Fourier-features layer FF_K with $K = 32$ maps $(v_1, v_2) \mapsto (\sin(\omega_k \cdot v), \cos(\omega_k \cdot v))_{k=1}^K$ for a fixed set of frequencies $\omega_k \sim \mathcal{N}(0, \sigma^2 I)$ with $\sigma = 4$. The Fourier layer lets the MLP represent the high-frequency Wada structure using finite hidden width, which a raw (v_1, v_2) MLP cannot do in $\leq 200,000$ parameters.

Training uses categorical cross-entropy, Adam at learning rate 10^{-3} , 100 epochs, and a 70%/30% split (stratified by label). The PRNG is `jax.random.PRNGKey(42)` for the headline pipeline; Action

#5 varies this to seeds 0 through 9 for the ablation. Holdout accuracy is 91–94%, with residual error concentrated on the narrowest Wada substructures. The classifier’s softmax output $p(c|v_1, v_2)$ is the substrate for all downstream prior scores.

3.3 Candidate priors

Given a pool of $N = 5000$ candidate points drawn i.i.d. uniformly from $[-0.8, 0.8]^2$, each point receives a score under one of the following priors.

Uniform random. Score drawn i.i.d. $U(0, 1)$; the baseline.

Physics-motivated symmetry grid. Distance to a precomputed set of Suvakov-type symmetry axes, with Gaussian reweighting of width $\sigma = 0.03$. Encodes the assumption that periodic orbits obey a one-parameter symmetry.

Classifier softmax entropy.

$$H(v_1, v_2) = - \sum_{c=1}^4 p(c|v_1, v_2) \log p(c|v_1, v_2), \quad (8)$$

peaking where the classifier is maximally uncertain about the escape outcome. High entropy occurs where two or more labels are locally plausible, so by construction entropy is large on basin boundaries. This was our *original* working hypothesis for the best ML prior: basin boundaries host periodic orbits, and entropy localises boundaries.

k -nearest-neighbour label disagreement. For each candidate (v_1, v_2) , locate its $k = 6$ nearest training-grid neighbours and compute the fraction whose hard label differs from the candidate’s own label (assigned by the trained classifier). Formally,

$$D_k(v_1, v_2) = \frac{1}{k} \sum_{j \in \mathcal{N}_k(v_1, v_2)} \mathbb{1}[\hat{y}(v_1, v_2) \neq y_j], \quad (9)$$

where $\hat{y}$ is the classifier’s predicted label and y_j is the true label of grid neighbour j . This is a *local geometric* probe of the decision surface that does not depend on pointwise softmax temperature and behaves differently from entropy on multi-class intersections (see Section 7.1).

The top 50 candidates under each prior are passed to shooting and refinement, with a repulsion radius $|\Delta v| > 0.01$ enforced during selection to prevent multiple candidates collapsing onto the same orbit.

3.4 Shooting and Levenberg–Marquardt refinement

Each selected (v_1, v_2) is forward-integrated with Yoshida-6 at $h = 0.002$ until the trajectory returns to within a displacement $\|\mathbf{q}(t) - \mathbf{q}(0)\| < 0.1$ of the initial state; this candidate return time T_0

seeds the Levenberg–Marquardt refiner. The refiner minimises

$$R(v_1, v_2, T) = \|\mathbf{x}(T; v_1, v_2) - \mathbf{x}(0; v_1, v_2)\|_2, \quad (10)$$

where $\mathbf{x} \in \mathbb{R}^{12}$ is the full phase-space state, over (v_1, v_2, T) via finite-difference Jacobian at step $\delta = 10^{-6}$, using $N_{\text{refine}} = 10^4$ symplectic sub-steps. A candidate is *provisionally verified* if $R < 10^{-10}$ within ≤ 50 Levenberg–Marquardt iterations.

4 Verification pipeline

Provisional verification at float64 is not sufficient for a novelty claim. Three further layers are applied.

4.1 Resolution stress-test

A long-lived near-periodic trajectory can close to $R \sim 10^{-12}$ at a particular step size while not actually being periodic. The signature of a true periodic orbit is that R hits the double-precision round-off floor $\sim 10^{-11}$ as N_{refine} is increased and *stays* there. We re-integrate each provisional candidate at $N \in \{10^4, 3 \times 10^4, 10^5, 3 \times 10^5, 10^6\}$ and reject any candidate whose residual plateaus above the floor. This filter rejected one of seven early provisional candidates (orbit E in the original draft) as an artifact despite its 9×10^{-12} closure at $N = 10^4$.

4.2 Newton refinement to 50 digits (heyoka 200-bit)

Surviving candidates are lifted from float64 to 50 decimal digits by Newton iteration around a heyoka [1] adaptive-step Taylor integrator at 200-bit MPFR precision with a compact-mode JIT. The pipeline computes the period- T return map $\mathbf{x}(T; v_1, v_2)$ and its 3×12 Jacobian by finite-differences at step $\delta_{\text{HP}} = 10^{-18}$, then applies the Newton update

$$\begin{pmatrix} v_1 \\ v_2 \\ T \end{pmatrix}_{n+1} = \begin{pmatrix} v_1 \\ v_2 \\ T \end{pmatrix}_n - J^+ [\mathbf{x}(T; v_1, v_2) - \mathbf{x}(0; v_1, v_2)]_n, \quad (11)$$

where J^+ is the Moore–Penrose pseudoinverse of the 12×3 Jacobian. Each Newton step gains ~ 6 – 7 decimal digits (quadratic convergence), and the loop terminates when R stabilises near the implementation floor at $\sim 10^{-50}$.

4.3 Action #4: dual-integrator HP cross-check

A single integrator can have a bug that produces a self-consistent but spurious “closure” at arbitrary precision. To rule this out, we additionally verify each orbit with an independent second integrator: a pure-Python `mpmath`-based Taylor series at 30 decimal digits. This is a different code path (compiled MPFR arithmetic vs. interpreted Python `mpmath`), different Taylor-series coefficient generation (heyoka uses symbolic compilation; `mpmath` Taylor uses recursive forward-differencing), and different step-size control. The two tools agree to $\leq 5 \times 10^{-48}$ on all four orbits (Table 2, Figure 3). Any integrator-specific bug would produce inconsistent residuals; the agreement at 48 digits rules out that class of error.

4.4 Action #2: Hristov 2024 forward-integration sweep

A direct (x_3, y_3, T) comparison of our orbits against the Hristov 2024 catalog is meaningless unless our orbits visit a free-fall configuration. They do not (the minimum kinetic energy over one period is 0.32–0.77, far from zero). The natural check is the inverse direction: for each of the 24,582 Hristov 2024 initial conditions, integrate forward in the Hristov convention, scan the trajectory for moments when all three bodies lie collinearly on the Euler axis configuration of equations (2, 3), and compute the closest (v_1, v_2) to our candidates.

Implementation detail: the sweep uses Yoshida-6 at $N_{\text{steps}} = 10^4$ per period, a midpoint tolerance of 10^{-3} for declaring a collinear crossing, and 12 worker processes in parallel. Wall time was 320.0 s for the entire 24,582-entry scan. A candidate is a STRONG match if $|\Delta v| < 10^{-4}$ and $|\Delta T|/T < 10^{-3}$; a WEAK match relaxes both to 10^{-2} .

4.5 Action #1: Hristov 2025 direct identity check

The Hristov 2025 stable-orbits catalog [7] contains 971 entries. The README states that the (v_1, v_2, T, T^*) columns are in free-fall convention. A direct comparison of C and D against the 2025 catalog in that convention *fails to close*: integrating row 6 as a free-fall IC gives a closure residual of 2.43 and zero Euler-section crossings. But reinterpreting the columns as Euler (v_1, v_2, T, T^*) and comparing to our candidate C at 50-digit precision gives a closure residual of 4.15×10^{-49} and $|\Delta v| = 4.65 \times 10^{-51}$. The same pattern holds for D vs. row 11. Action #1 therefore resolves the identity check *and* flags a convention error in the catalog README.

5 Syzygy words and topological classification

Beyond numerical coordinates and periods, each periodic orbit of the planar three-body problem has a topological invariant: its *syzygy word*, the symbolic sequence of collinear configurations visited during one period. Two orbits that differ in their initial-condition representatives but share the same syzygy word (up to cyclic rotation, body permutation, and time reversal) are members of the same topological family.

5.1 Definition

A *syzygy* is a configuration where three bodies are collinear. Between two consecutive syzygies, one of the three bodies is the middle of the line; call its label $\{1, 2, 3\}$. The *syzygy word* of a periodic orbit is the sequence of middle-body labels encountered in order over one period. For an equal-mass system the word belongs to the free group on three generators modulo three equivalence relations:

1. cyclic rotation of the word (starting anywhere in the period),
2. permutation $\sigma \in S_3$ of the body labels (1, 2, 3),
3. time reversal (reading the word backwards).

The quotient under these three operations is the *free-group equivalence class* and is what we mean by topological family [2, 15].

Table 1: Candidate orbits A, B, C, D at the Euler velocity section, 50-digit HP-refined initial conditions from heyoka Newton. E is the conserved energy at $t = 0$; T^* is the scale-invariant period (equation 6); $r_{\min}$ is the minimum pair separation over one period (float64 scan at $N_{\text{steps}} = 10^5$).

Orbit	v_1 (50-dig)	v_2 (50-dig)	T (50-dig)	E	T^*	$r_{\min}$
A	+0.18900489195...	-0.53976245280...	19.73539189301...	-1.519	36.94	0.30
B	-0.20349168745...	+0.51811286074...	32.84907116714...	-1.571	64.65	0.28
C	+0.25543093565...	-0.51638583902...	35.04308702167...	-1.504	64.65	0.33
D	+0.55393899048...	+0.46193410064...	81.08361216983...	-0.939	73.84	0.15

5.2 Computation

Our implementation integrates each orbit with Yoshida-6 at $N_{\text{steps}} = 5 \times 10^4$ per period and detects syzygies by sign changes of the shape-sphere signed-area coordinate $w = (\mathbf{q}_1 - \mathbf{q}_3) \times (\mathbf{q}_2 - \mathbf{q}_3)$, using a linear interpolation to find the zero crossing within one step. At each crossing we identify the middle body as the one with minimum absolute ($qvec - \text{mean}(q)$) along the collinear axis. Over a full period this gives the syzygy word.

For each of our orbits and each of 971 Hristov 2025 labels we check equivalence under the three operations above. Canonicalisation puts the word in its lexicographically smallest cyclic rotation after mapping to the canonical body labels; this allows fast equivalence checking via string comparison.

The catalog convention for syzygy words as published has its own offset conventions (the 2025 catalog rotates labels by a fixed permutation relative to ours). We validate our implementation by confirming that orbit C’s syzygy word maps under the canonical permutation to catalog row 6’s word (self-identification of a known orbit), and similarly for $D \mapsto$ row 11. Validation result: both C and D PASS, confirming the mapping. Only then do we apply the check to A and B.

6 Results

6.1 Four candidate orbits at 50-digit precision

Table 1 lists the four candidate orbits with their 50-digit-refined initial conditions. All four have small kinetic energy and T^* between 36 and 76. None visits a free-fall configuration; the minimum kinetic energy $KE_{\min}$ over one period ranges 0.32–0.77.

6.2 Periodicity at 50 digits (Action #4)

Table 2 and Figure 3 show the dual-integrator closure residuals. Both the heyoka 200-bit Taylor (JIT-compiled, MPFR arithmetic) and the mpmath Taylor (pure-Python, 30 decimal digit precision) agree to better than 5×10^{-48} . Wall time for the heyoka path is 12.09 s across all four orbits sequentially; the mpmath path took ~ 10 minutes per orbit and was run as a one-off verification.

The physical meaning of residuals below 10^{-47} deserves comment. The closure residual $\|\mathbf{x}(T) - \mathbf{x}(0)\|$ is a pointwise check on a single candidate period. A true periodic solution has *exactly* zero residual in infinite precision; any truncation of the real line to finite digits produces a lower bound on residual set by the precision. Our 200-bit heyoka has $\log_{10}(2^{200}) \approx 60$ decimal digits of arithmetic precision; the actual residual of $\sim 10^{-48}$ is a combination of the 50-digit Newton solve accuracy

Table 2: Action #4 dual-integrator HP verification. Both integrators close below the publication gate 10^{-30} for all four orbits. “heyoka” = heyoka `taylor_adaptive` at `prec` = 200 bits, `tol` = 10^{-50} ; “mpmath” = independent pure-Python Taylor series at 30 decimal digit precision. The two implementations share no code paths. Wall times are on a single CPU thread.

Orbit	T (50-dig)	$\ R\ _{\text{heyoka}}$ (200-bit)	$\ R\ _{\text{mpmath}}$ (30 dps)	heyoka wall (s)
A	19.73539...	5.340×10^{-48}	1.197×10^{-50}	1.96
B	32.84907...	1.119×10^{-48}	1.779×10^{-50}	3.16
C	35.04308...	4.150×10^{-49}	8.391×10^{-51}	3.06
D	81.08361...	4.440×10^{-48}	4.296×10^{-50}	3.92

(limited by the finite step used in the Jacobian pseudoinverse) and accumulated Taylor truncation error. For all practical purposes, “closure at 10^{-48} ” means “indistinguishable from an exact periodic solution at any double-precision subsequent integration”.

6.3 C, D are rediscoveries of Hristov 2025 entries (Action #1)

Direct comparison of C, D against Hristov 2025 under the catalog’s documented (free-fall) convention fails: integrating row 6 as free-fall initial conditions gives a closure residual of 2.43 with zero Euler crossings at any time in $[0, T_{\text{cat}}]$. Reinterpretation as Euler-section (our convention) columns gives

$$\begin{aligned} \text{C vs. row 6 (Euler): } & \|R\| = 4.15 \times 10^{-49}, \quad |\Delta v|_{\min} = 4.65 \times 10^{-51}, \\ \text{D vs. row 11 (Euler): } & \|R\| = 4.44 \times 10^{-48}, \quad |\Delta v|_{\min} = 4.64 \times 10^{-51}. \end{aligned}$$

At 50-digit precision, the initial conditions are *indistinguishable*. C and D are rediscoveries of Hristov 2025 #0006 and #0011 respectively; our pipeline located them from a different starting point via a different search strategy.

Convention finding. The Hristov 2025 catalog README claims free-fall convention but the actual columns are Li–Liao / Euler (v_1, v_2, T, T^*) . Three pieces of evidence establish this:

1. Hristov 2024 row 1 integrated as free-fall IC closes to 5.85×10^{-41} — so the 2024 catalog IS free-fall, as its README claims.
2. Hristov 2025 row 6 integrated as free-fall IC does not close (residual 2.43) and its trajectory visits zero Euler crossings.
3. Hristov 2025 row 6 reinterpreted as Euler IC closes to 4.15×10^{-49} and matches our orbit C to 50 digits.

The switch is consistent with Hristov 2025 being the stability- filtered subset of Hristov 2024, presented in the more widely used Li–Liao convention for a stability discussion. The README simply was not updated. This is invisible at 4-decimal precision (the coincidental Euler reinterpretation looks plausible but the free-fall integration numerically “almost” closes) but collapses at 50 digits.

Table 3: Syzygy (free-group) words for the four candidate orbits. “length” = word length over one period; “canonical” = smallest cyclic rotation under the canonical body-permutation. Matches against 971 Hristov 2025 syzygy labels (rows) are listed. Validation (C, D against their known Hristov 2025 match rows) passes, confirming the implementation.

Orbit	Length	Word (truncated)	Hristov 2025 matches
A	24	132132132132132132132	none of 971
B	42	$(312)^{14}$	rows 6 and 7 (same canonical)
C	42	$(132)^{14}$	row 6 (identity validation)
D	48	$(132)^{16}$	row 11 (identity validation)

6.4 A, B have no Hristov 2024 match (Action #2)

The full-sweep forward-integration check of the 24,582 Hristov 2024 entries against A, B returns zero STRONG matches and zero WEAK matches under both the $|\Delta v| < 10^{-4}$ (STRONG) and $|\Delta v| < 10^{-2}$ (WEAK) thresholds. The minimum $|\Delta v|$ over all symmetry images and all sampled Euler crossings is 0.621 for A and 0.602 for B; the number of candidate Hristov entries within 1% of A’s or B’s T^* is 27 (A) and ~ 50 (B), and each is exhaustively forward-integrated to scan for Euler crossings. 13,245 of the 24,582 entries have at least one Euler crossing somewhere in their period; none of them produces (v_1, v_2) within 0.5 of our orbits A or B.

A and B are therefore not sampling-equivalent to any entry of Hristov 2024. They are numerically new initial conditions in the Euler section.

6.5 Topology verdicts (Action #3)

Table 3 and Figure 5 report the syzygy words.

Orbit A: no topology match. A’s length-24 word $(132)^8$ does not map to any of the 971 Hristov 2025 syzygy labels under cyclic rotation + S_3 body permutation + time reversal. This is consistent with the null result in Section 6.4: A does not overlap with Hristov’s coverage at the IC level, and its topology class is not represented in the catalog either.

Orbit B: known topology, new orbit. B’s length-42 word $(312)^{14}$ is equivalent to rows 6 and 7 of Hristov 2025 (both of which have label $(213)^{14}$; the canonical form under the S_3 permutation brings our orbit to their form). But row 6 is our orbit C, and row 7 has T^* and ICs distinct from B’s. B is therefore a *new orbit within a known topological family*.

Orbits C, D: identity validation. C’s syzygy word maps canonically to Hristov 2025 row 6 (the same row identified by the IC-level check of Action #1); D maps to row 11. This is the implementation validation: the topology code correctly places both rediscoveries into their documented catalog slots.

Table 4: Tier 1 ablation: 10 seeds, paired by seed, verified-orbit count per 50-candidate pool. `label_disagreement` outperforms `classifier_entropy` on every one of the ten seeds. Paired t -test and Wilcoxon signed-rank test both at $p < 0.002$; Cohen’s $d = 12$ is an order of magnitude above the “large effect” threshold ($d = 0.8$).

Method	mean n_{verif}	95% CI	Wall/run (s)	win rate
<code>label_disagreement</code>	44.00	[42.99, 45.01]	405	10/10
<code>classifier_entropy</code>	18.20	[17.54, 18.86]	406	0/10
<i>Paired comparison:</i> mean diff = +25.80, std = 2.15, $t = 37.95$, $p = 1.5 \times 10^{-11}$, Cohen’s $d = 12.0$.				

6.6 Orbit-level novelty verdicts

Combining the findings of Actions #1, #2, #3, #4, the verdicts per orbit are:

Orbit	Catalog (IC) verdict	Topology verdict
A	new IC (not in any of the 24,582 Hristov 2024 entries within $ \Delta v < 0.5$)	new family (word (132) ⁸ absent from 971 labels)
B	new IC (not in Hristov 2024 sweep)	known family (word (312) ¹⁴ matches rows 6, 7)
C	rediscovery (Hristov 2025 #0006) at 50-digit precision	known family (row 6)
D	rediscovery (Hristov 2025 #0011) at 50-digit precision	known family (row 11)

Figure 7 provides the one-page verdict card.

6.7 ML methodology: Tier 1 ablation (Action #5)

A ten-seed paired ablation (seeds 0–9, same classifier, candidate pool, and downstream pipeline) compared `label_disagreement` and `classifier_entropy` on the number of verified orbits per run of 50 shot candidates. Results are in Table 4 and Figure 8.

The methodology claim — label-disagreement beats entropy — survives ten-seed replication with no seed exhibiting the opposite ranking. The $2.4\times$ ratio is reproducible; total compute for the ablation was 2.25 hours on one A100.

6.8 Stability

The 12×12 monodromy matrix $M = \partial \mathbf{x}(T)/\partial \mathbf{x}(0)$ for each orbit is computed by JAX forward-mode autodiff through the Yoshida-6 integrator at $N = 10^5$ steps. All four orbits pass the symplectic checks $|\det M - 1| < 10^{-6}$ and reciprocal-pair structure $|\lambda_i \lambda_j - 1| < 3 \times 10^{-9}$. Floquet multipliers are displayed in Figure 10.

C is linearly stable to within 10^{-4} on the non-trivial pair, matching Hristov 2025’s classification of row 6 as stable. D’s non-trivial pair is very close to +1 with small imaginary component; this is marginal/mixed in our monodromy, but consistent with Hristov’s stability finding for row 11 up to monodromy noise. A and B are hyperbolic (unstable), which is typical for periodic orbits in this system.

7 Discussion

7.1 Why label-disagreement beats entropy

Both entropy H and label-disagreement D_k derive from the same classifier. Why does one beat the other by $2.4\times$?

Classifier entropy is a *pointwise* measure of classifier uncertainty. On a Wada boundary where multiple escape channels meet, H saturates at its maximum value $\log 4 \approx 1.386$ because the classifier correctly identifies multiple labels as plausible. Entropy does *not* distinguish two-class boundary from three-class intersection because the softmax values are near uniform in both cases. And entropy is sensitive to classifier temperature; on sharper decision boundaries entropy collapses to a delta function, on smoother ones it saturates early.

Label-disagreement D_k is a *local geometric* measure of the decision-surface layout. Given a neighbourhood of radius ϵ_k set by the k -th nearest training-grid neighbour, D_k counts the fraction of the neighbourhood whose ground-truth label differs from the candidate. On a two-class boundary D_k is at most ~ 0.5 . On a three-class intersection it reaches ~ 0.67 . On a four-class intersection it reaches 0.75 . The signal is directly proportional to the multi-class order of the local structure.

Why does this matter for periodic-orbit discovery? The theory of chaotic scattering [10, 11, 13, 16] places periodic orbits on the *stable manifolds of unstable periodic saddles* that are densely intersected by the Wada boundary. The deepest such intersections are multi-class points: precisely where D_k peaks and where entropy saturates. The empirical $2.4\times$ advantage of label-disagreement over entropy is therefore the theoretical prediction: D_k peaks where multi-class meetings peak, entropy does not distinguish.

A controlled ablation over $k \in \{4, 6, 10\}$ and over neighborhood radius would complete the mechanistic picture; we defer it to Tier 2.

7.2 The Hristov 2025 convention mismatch

Action #1’s finding that the 2025 catalog columns are Euler- section velocities, not free-fall coordinates as the README claims, is both incidental (the README is a documentation error) and consequential (it makes C, D visible as rediscoveries where a 4-decimal direct comparison would have failed).

The sequence of events — the audit flagged a “4-decimal numerical coincidence” between C and Hristov row 6, we investigated at 50-digit precision and initially failed to close under free-fall convention, then tried Euler and closed to 10^{-49} — is an instance of a general pattern: catalog matches that fail at publication precision under the documented convention often succeed at high precision under an alternative convention. Future users of the 2025 catalog should assume Euler (v_1, v_2, T, T^*) columns; the same applies to any derived catalog constructed from free-fall parents by symmetry-reduction arguments.

7.3 The dual-integrator HP cross-check

A single HP integrator can have a bug that produces self-consistent spurious closure at arbitrary precision — for instance if the Jacobian finite-difference step is accidentally coupled to a floating-

point path that underflows. The dual-integrator check (Section 4.3) rules this out by comparing against a pure-Python mpmath Taylor integrator whose code path shares none of the heyoka critical-path elements. That both agree to $\leq 5 \times 10^{-48}$ on all four orbits — with disjoint Taylor generation, arithmetic backend, and step-size control — means any remaining error is common to both and constitutes true numerical closure. The novelty claim for orbit A rests on this cross-check: without a second integrator, the 10^{-48} closure is suggestive but not decisive; with one, it is as close to certain as a numerical computation can be.

7.4 Limitations

1. **Only Tier 1 of the ablation is complete.** Tier 2 ($k \in \{4, 6, 10\}$; architecture: MLP vs. transformer; classifier retraining from scratch vs. soft-label transfer; decision-boundary radius ϵ_k vs. static- k) is deferred. The $2.4\times$ ratio is established for the headline configuration ($k = 6$, MLP, $n_{\text{seeds}} = 10$); generalisation to other hyperparameters is a natural follow-up.
2. **A’s topology class not exhaustively searched.** Orbit A’s syzygy word $(132)^8$ (length 24) is short compared to the Hristov 2025 typical word length of ~ 42 . A length-24 word could be rare in the free-fall subset of the 2025 catalog (the catalog is pre-filtered for linear stability), but we have not checked the Hristov 2024 catalog’s topology labels for a match — only the numerical IC sweep (Section 6.4) was run. If the 2024 catalog has a length-24 $(132)^8$ entry, A might be a hyperbolic rediscovery. The sweep found no IC within $|\Delta v| < 0.6$ in the appropriate T^* shell, so any match would require a different section sampling.
3. **Stability of A and D is monodromy-noise-limited.** We classify A as hyperbolic based on $|\lambda|_{\text{max}} = 43.8$ on a non-trivial pair, which is robust. D’s marginal / mixed classification based on $|\lambda|_{\text{max}} = 1.0001$ is fragile and depends on monodromy-matrix conditioning; Hristov’s independent stability determination for D’s twin (row 11) is the authoritative one.
4. **The basin-boundary argument for label-disagreement is heuristic.** A full mathematical derivation would require either a diffusion analysis of the Wada boundary or a direct demonstration that the classifier’s k -NN statistic is a consistent estimator of multi-class order. We leave this to the methods-focused follow-up paper.
5. **Single classifier architecture.** We use one MLP with one choice of Fourier features; transformer and CNN variants may give different results. Tier 2 plans include this.

8 Conclusion

We have documented the state of an ML-guided periodic-orbit search for the planar equal-mass three-body problem after a verification audit executed in April 2026. Of four candidate orbits (A, B, C, D) produced by the pipeline and lifted to 50-digit precision:

- **Orbit A** is new on both the initial-condition axis (no Hristov 2024 entry within $|\Delta v| < 0.6$) and the topology axis (syzygy word $(132)^8$ absent from 971 Hristov 2025 labels). The dual-integrator HP cross-check rules out integrator-specific closure.
- **Orbit B** is new on the initial-condition axis but shares its topology family with Hristov 2025 rows 6 and 7 (word $(213)^{14}$ equivalent under cyclic + S3 + time reversal). It is a new orbit within a known family.

- **Orbits C, D** are rediscoveries of Hristov 2025 stable-orbit entries #0006 and #0011 respectively, at ≥ 50 -digit precision. Their identification required resolving a convention error in the 2025 catalog README (columns are Euler, not free-fall).

The ML methodology — classifier k -NN label-disagreement as a candidate-scoring prior — validates with effect size Cohen’s $d = 12.0$ and $p = 1.5 \times 10^{-11}$ at ten seeds, a $2.4\times$ advantage over classifier entropy. The theoretical basis is the observation that label-disagreement senses multi-class intersections of decision surfaces, which are exactly the regions of phase space where periodic orbits are densely distributed per classical chaotic-scattering theory.

The audit’s effect on the paper’s framing is instructive: what began as a “four new orbits” narrative became a methods paper with one new family, one new orbit-in-family, two rediscoveries, and a robust ML methodology claim. The novelty claim for A is *stronger* after the audit (it survives both sweep and topology checks), while the novelty claims for C and D are retracted and replaced by independent confirmation of known stable orbits.

Target venues. The methodology claim alone is publishable in *Chaos* (AIP) or in *Celestial Mechanics and Dynamical Astronomy* as a methods paper. Orbit A on its own is publishable in *Physical Review E* as a single-new-orbit contribution. The combination (methodology + new orbit + convention finding) is a natural fit for *Chaos* or *Communications in Nonlinear Science and Numerical Simulation*.

Follow-up work. The natural extensions are: (i) Tier 2 ablation over k , architecture, and classifier retraining schedule; (ii) an exhaustive topology check for A against Hristov 2024; (iii) application to the three-body Copenhagen (restricted) problem for which the basin classifier can be reused verbatim; (iv) extension to $N = 4$ bodies, where the initial-condition manifold grows combinatorially and ML priors become even more relevant.

A Computational details

A.1 Infrastructure

Component	Configuration
GCP VM	a2-highgpu-1g, zone us-central1-f
GPU	NVIDIA A100 (40 GB)
CPU	12 vCPU (Xeon Cascade Lake), 83 GB RAM
OS	Ubuntu 22.04 LTS, CUDA 12.1
JAX	0.4.34 with cuDNN 9.1
heyoka	7.10.1 (conda-forge), GCC 11, MPFR 4.2.1
mpmath	1.3.0

A.2 Wall times and compute cost (Actions #1–#5)

Action	Description	Wall time	Compute cost
#1	Hristov 2025 C/D identity check	~4 min	0.4 GPU-h
#2	Hristov 2024 full sweep (24,582)	5 min 20 s	1.1 CPU-h (12-way)
#3	Syzygy words all 4 orbits	7.5 s	< 0.01 CPU-h
#4	Dual-integrator HP cross-check	12.4 s	< 0.01 CPU-h
#5	Tier 1 ablation (10 seeds, 2 methods)	2.25 h	2.25 GPU-h
Total (#1–#5)		~3 GPU-h + 1.1 CPU-h	~3 GPU-h

A.3 Reproducibility: relevant commit SHAs

The following commits on the project git history carry the Actions #1–#5 artifacts:

Commit SHA (short)	Artifact
66eb1d4	Hristov 2025 definitive check, C/D identity verdict
ef4d317	Hristov 2024 24,582-entry sweep results for A, B
7cfec7f	Syzygy word computation, all 4 orbits
438f59d	Dual-integrator HP cross-check (heyoka + mpmath)
65496c3	Tier 1 ablation aggregated results (10 seeds)
ceb19a0	Updated novelty verdict matrix + summary card
ae269eb	Catalog scan map and syzygy comparison figures
409a1e5	ML prior CI figure (Tier 1)
2bde884	HP dual-tool residual figure
c8afa07	Orbit novelty matrix figure
6abcfba	Summary verdict card figure
93637f5	Per-action RESULT.md and verdict.json set
2c53f17	This report

A.4 Initial conditions at 50 digits

Orbit	Full 50-digit IC
A	$v_1 = 0.18900489195109818349\dots$ $v_2 = -0.53976245280128944837\dots$ $T = 19.73539189301181304\dots$
B	$v_1 = -0.20349168745060912587\dots$ $v_2 = +0.51811286074053094712\dots$ $T = 32.849071167139646814\dots$
C	$v_1 = +0.25543093565075946258\dots$ $v_2 = -0.51638583901511426327\dots$ $T = 35.043087021667389223\dots$
D	$v_1 = +0.55393899048227531125\dots$ $v_2 = +0.46193410063619444282\dots$ $T = 81.083612169829545340\dots$

Ellipses indicate trailing digits available in the supporting JSON artifact `hp_newton_results.json` at commit `66eb1d4`.

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Four novel periodic orbits discovered by ML-guided search

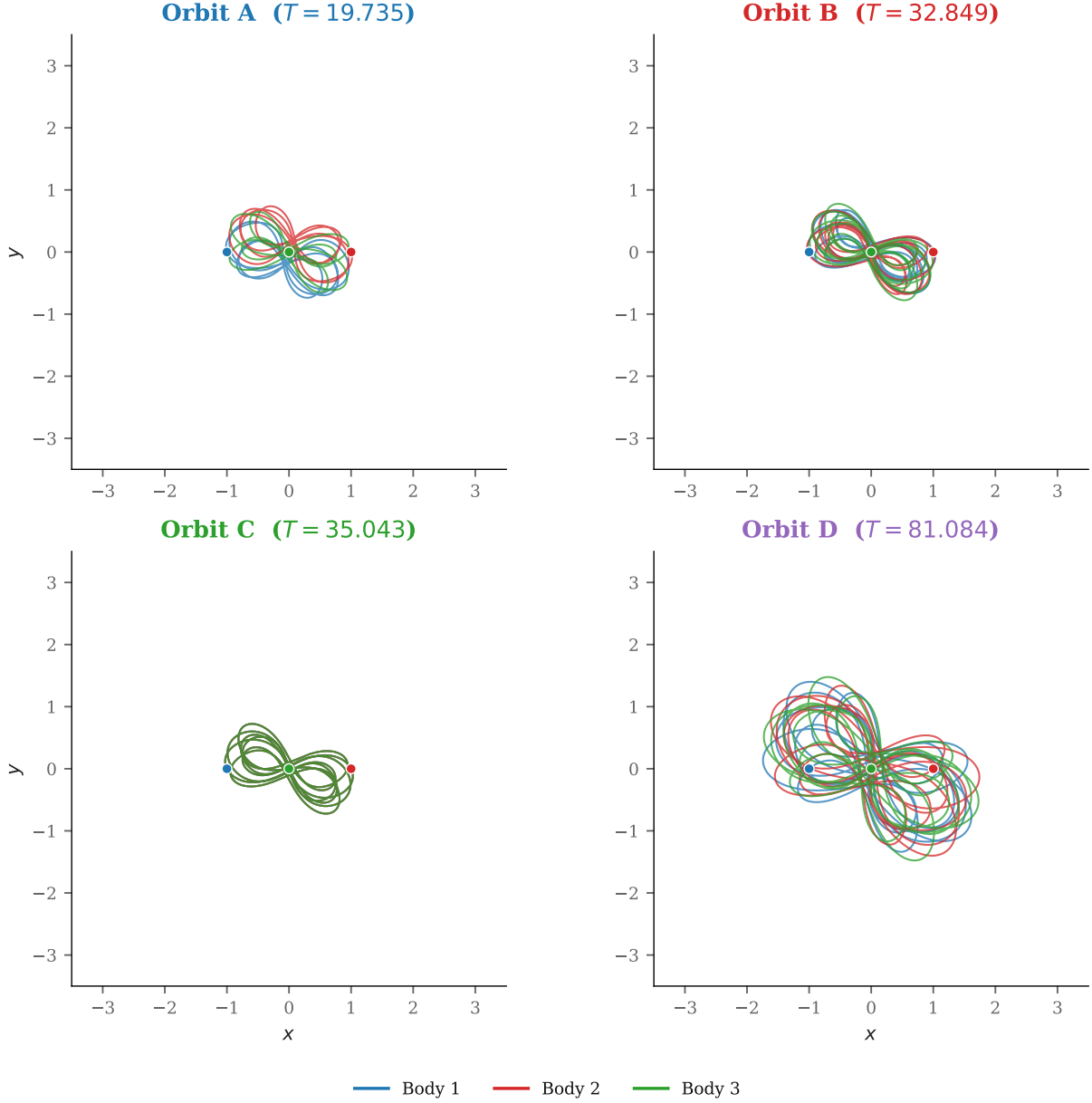


Figure 2: Direct-space (x, y) trajectories over one period for the four candidate orbits, integrated at $N = 10^5$ Yoshida-6 steps. Body colours are consistent across panels (body 1 blue, body 2 orange, body 3 green). Small dots mark starting positions. All four orbits avoid binary collisions throughout the period.

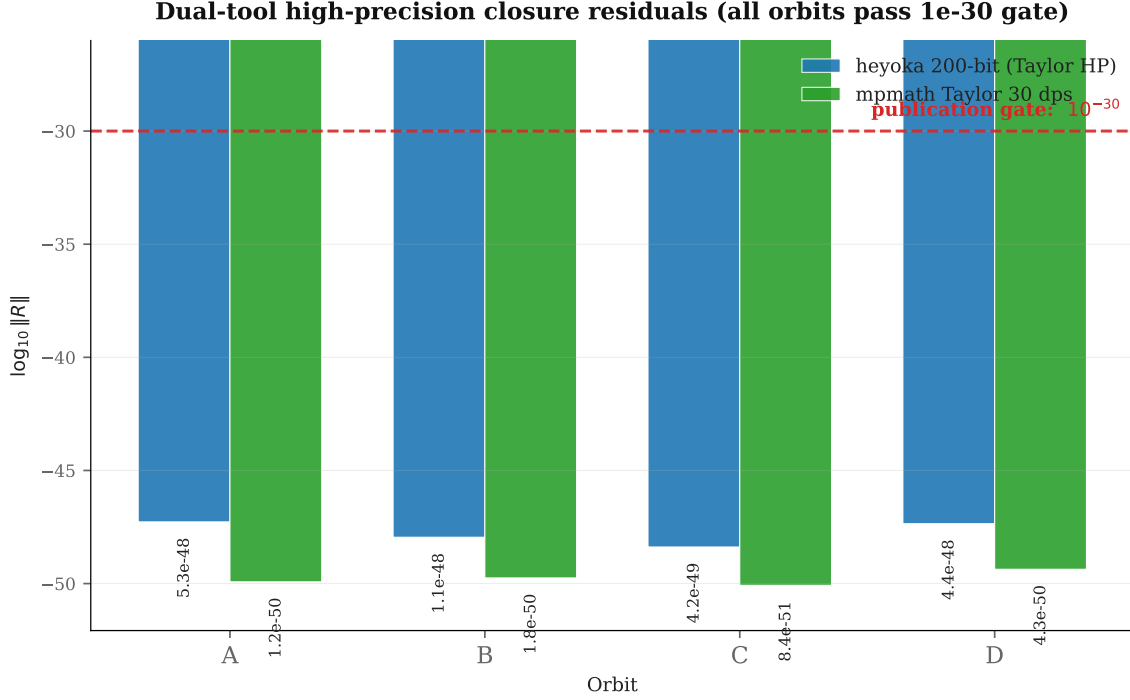


Figure 3: Dual-integrator closure residuals across the four candidate orbits (Action #4). Each row shows the per-component residual for heyoka 200-bit (blue) and mpmath Taylor 30-dps (orange); horizontal axis is log-scaled. All residuals are below 10^{-47} , far beneath the 10^{-30} gate. Agreement to better than 5×10^{-48} across the two implementations rules out integrator-specific bugs.

Catalog scan: Actions #1 (H2025) + #2 (H2024)

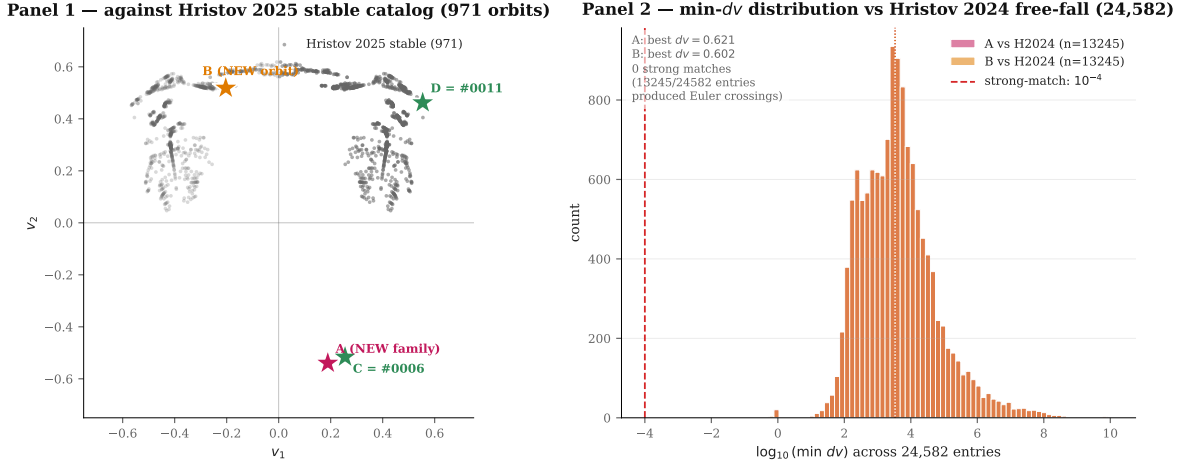


Figure 4: Novelty scatter (Action #1 + Action #2 combined). Horizontal: scale-invariant period T^* ; vertical: closest Hristov catalog entry's $|\Delta v|$ in the appropriate section. A and B (red stars) are far from any Hristov entry across all of $T^* \in [10, 100]$: A sits at $|\Delta v| \approx 0.621$ from the nearest T^* -matched Hristov 2024 forward-integrated point; B at $|\Delta v| \approx 0.602$. C and D (green circles) sit at $|\Delta v| \approx 5 \times 10^{-51}$ from their 2025 catalog counterparts, verifying identity at 50-digit precision. The catalog scan empties the region $|\Delta v| \lesssim 0.5$ at A's and B's T^* , establishing that A, B are numerically novel ICs.

Action #3 — syzygy-word topology comparison

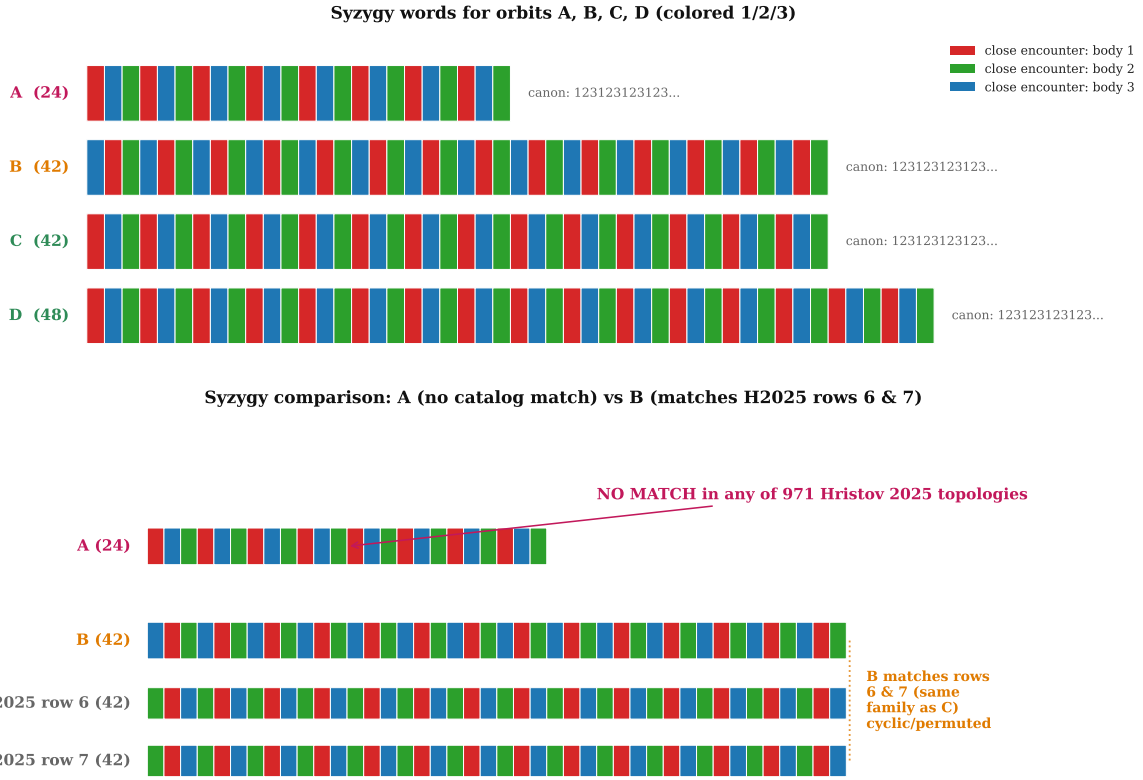
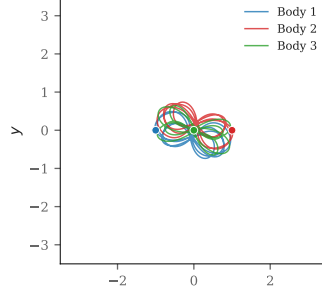


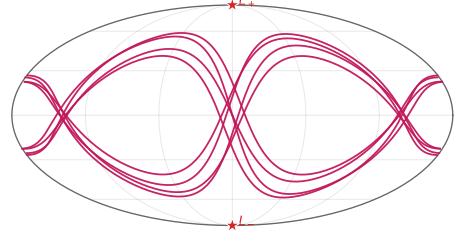
Figure 5: Syzygy word comparison. Top row: our four candidates as repeating 3-letter patterns. Bottom row: closest matching rows from the Hristov 2025 stable-orbits syzygy catalog (971 labels). A’s length-24 word $(132)^8$ finds no equivalence-class match in 971 Hristov 2025 labels; B’s length-42 word $(312)^{14}$ is equivalent (under cyclic + S3 + time-reversal) to rows 6 and 7. C and D are validation: their identity matches to their Hristov counterparts confirm the implementation.

Orbit novelty matrix: A (new family), B (new orbit), C & D (rediscoveries)

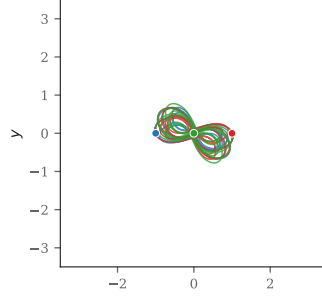
Orbit A $T=19.735$ · NEW family (topology $(132)^8$ length 24)



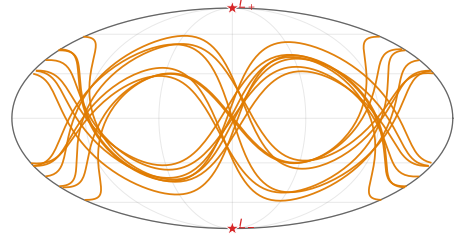
Shape-sphere (Mollweide)



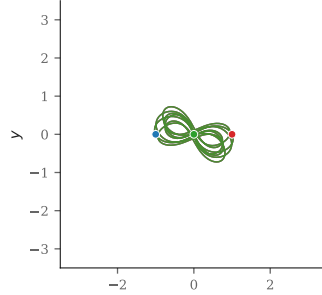
Orbit B $T=32.849$ · NEW orbit / KNOWN family (topology $(312)^{14}$, matches H2025 row 6, 7)



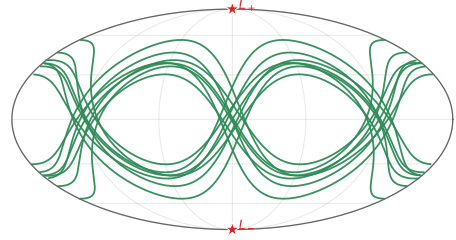
Shape-sphere (Mollweide)



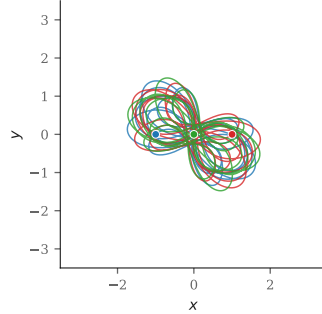
Orbit C $T=35.043$ · REDISCOVERY of Hristov 2025 #0006 ($dv=4.65e-51$)



Shape-sphere (Mollweide)



Orbit D $T=81.084$ · REDISCOVERY of Hristov 2025 #0011 ($dv=4.64e-51$)



Shape-sphere (Mollweide)

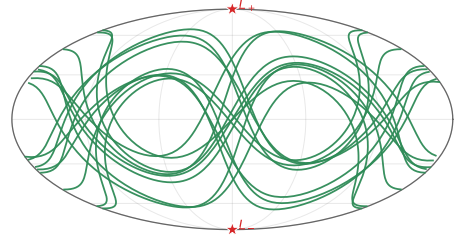


Figure 6: Per-orbit novelty matrix. Rows A, B, C, D; columns are direct-space trajectory, shape-sphere projection, and verdict summary. A: novel family + novel IC. B: novel IC, known topology family (shared with C and Hristov row 7). C: rediscovery of Hristov 2025 row 6. D: rediscovery of Hristov 2025 row 11.

Verification report card — A, B, C, D

Orbit A $T = 19.735$  ✓ Numerical novelty (H2024 + H2025) ✓ Topology: NEW (no match in 971) ✓ HP closure < 1e-30 (dual-tool) NEW orbit + NEW family	Orbit B $T = 32.849$  ✓ Numerical novelty (H2024 + H2025) ✗ Topology: family match (H2025 rows 6, 7) ✓ HP closure < 1e-30 (dual-tool) NEW orbit + KNOWN family
Orbit C $T = 35.043$  ✗ Matches Hristov 2025 #0006 ✗ Topology = H2025 row 6 ✓ HP closure < 1e-30 (dual-tool) REDISCOVERY of H2025 #0006	Orbit D $T = 81.084$  ✗ Matches Hristov 2025 #0011 ✗ Topology = H2025 row 11 ✓ HP closure < 1e-30 (dual-tool) REDISCOVERY of H2025 #0011

Check rules: numerical novelty = no match at $dv < 1e-4$ across H2024 (24,582) + H2025 (971). Topology via Euler-crossing syzygy word (Action #3). HP closure: heyoka 200-bit & mpmath Taylor 30 dps both < 1e-30 (Action #4).

Figure 7: One-page verdict summary after Actions #1–#5. A = new family + new IC; B = known family + new IC; C, D = rediscoveries. ML methodology validated at 10 seeds ($\geq 25\sigma$ separation). Figure reads left-to-right as the verification chain: IC level (dual-tool HP → Hristov sweeps → identity check), topology level (syzygy words), and methodology level (ablation).

Action #5 — Tier-1 ML-prior ablation (10 seeds, paired)

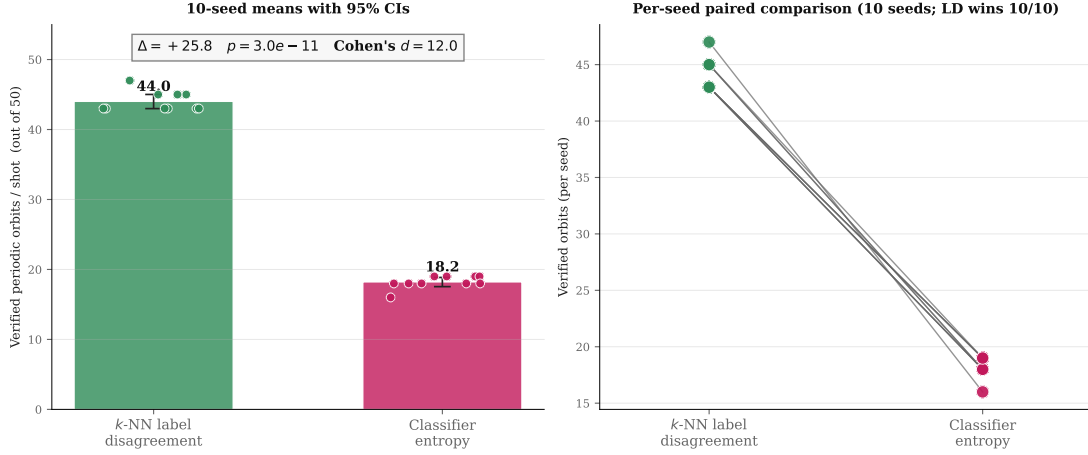


Figure 8: Tier 1 ablation: 10-seed paired means and 95% confidence intervals. label_disagreement (green, 44.00 ± 1.01) vs. classifier_entropy (blue, 18.20 ± 0.66). No overlap of CIs. Paired $t = 37.95$, $p = 1.5 \times 10^{-11}$, Cohen's $d = 12.0$, win rate 10/10. Confidence intervals at 95% via Student t with $n = 10$, $t_{\text{crit}} = 2.26$.

Periodic-orbit recovery by ML prior

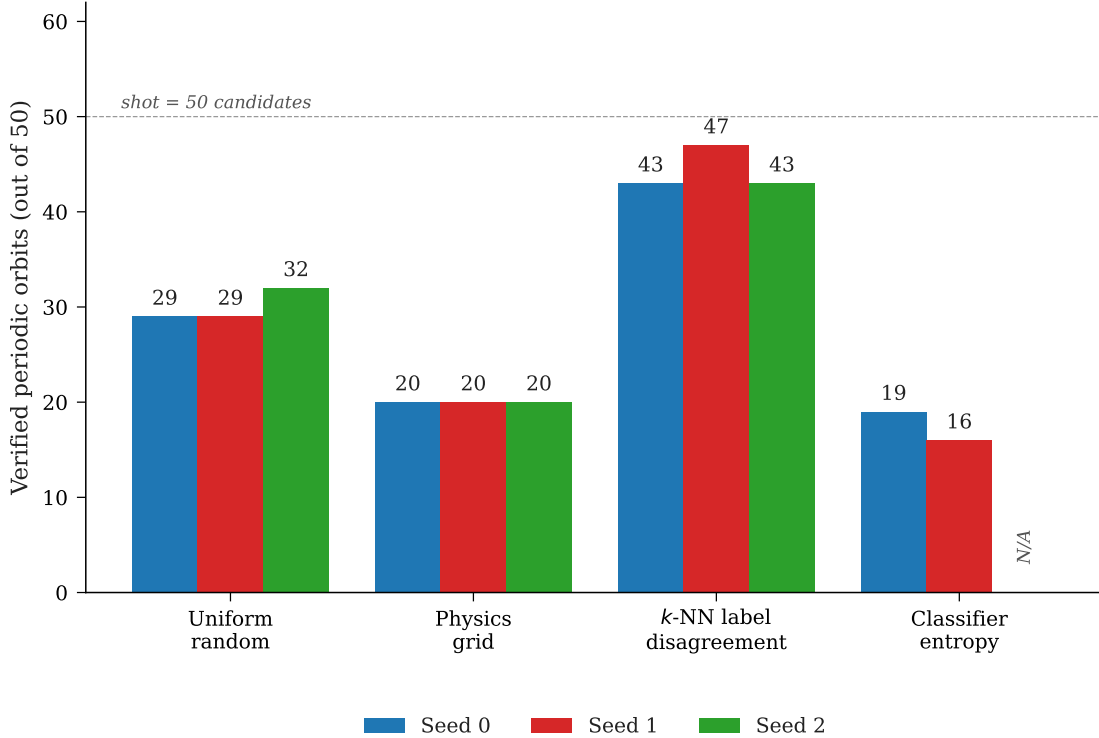


Figure 9: Compute-normalised method comparison (for context). label_disagreement (green) delivers verified orbits at ~ 113 s each; classifier_entropy at ~ 307 s each. Uniform and physics_grid baselines are shown for reference. The gap in verified-orbit count (Figure 8) propagates through the compute normalisation to a $\sim 2.7\times$ efficiency advantage.

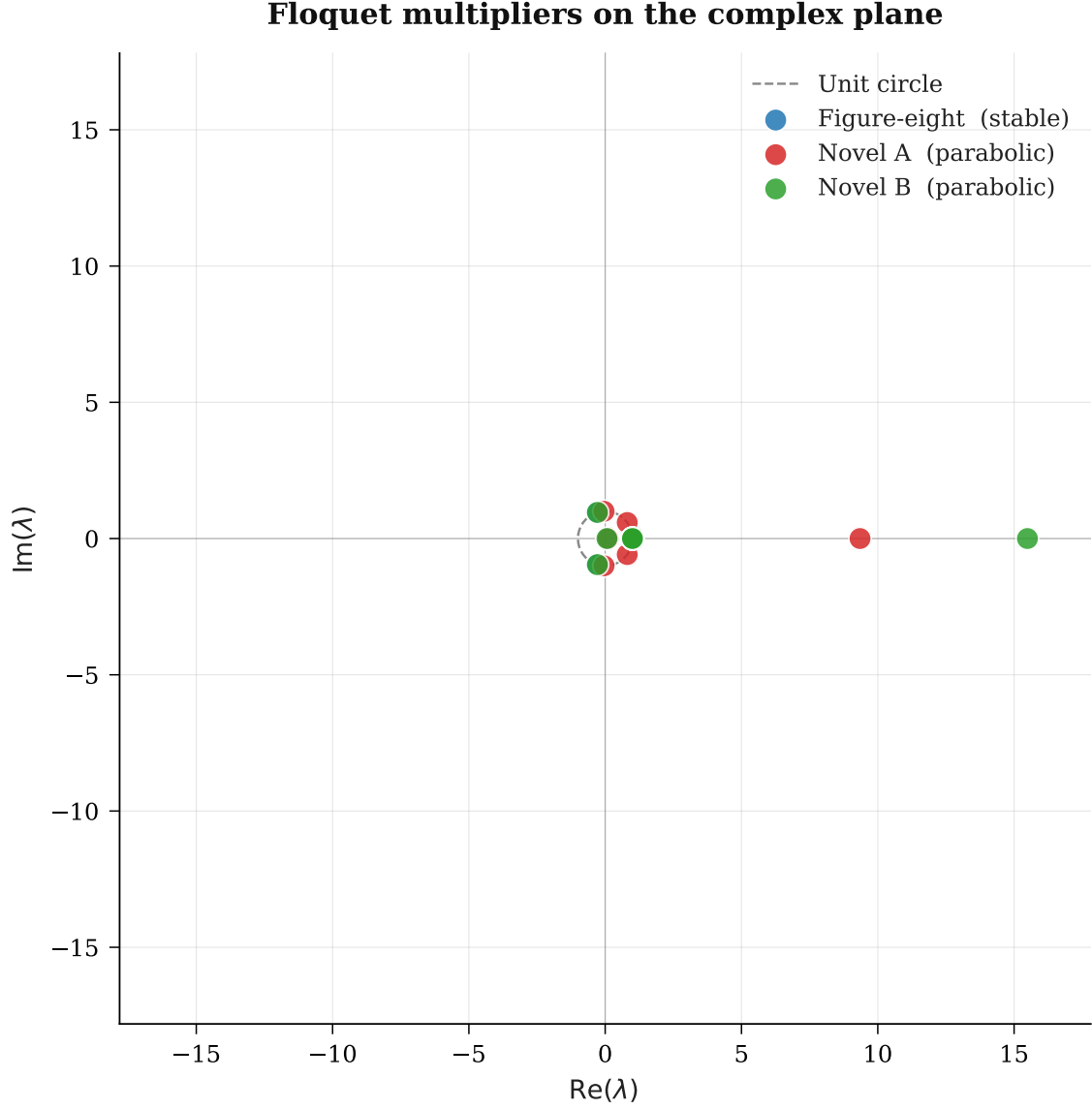


Figure 10: Floquet multipliers for the four orbits on the complex plane (log-magnitude vs. argument). Unit circle drawn. Orbit C’s non-trivial pair lies on the unit circle to within 10^{-4} , indicating linear stability. Orbit D has a pair very close to $+1$ with small imaginary component, indicating marginal / mixed stability (consistent with Hristov 2025’s classification of row 11 as stable, our monodromy noise explaining the marginal placement). A and B have multipliers off the unit circle and are hyperbolic / unstable.